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Combinatorics – II

Derangements

It is an arrangement of objects in which no object is in its original position.

Eg:- A permutation of integers 1, 2, 3, ... n in which 1 is not in the first place, 2 is not in the second place, 3 is not in the third place and so on and n is not in the n^{th} place is a derangement.

The no. of possible derangements of n distinct objects 1, 2, 3, ... n is denoted by d_n. If there is only one object it continues to be in its original place in every arrangement. then $d_1 = 0$. If there are two objects, a derangement can be done in only one way by interchanging their places, then $d_2 = 1$. For three objects, 1, 2, 3, the possible derangements 2, 3, 1 & 3, 1, 2 then $d_3 = 2$. In general, the no. of derangements of n distinct objects is

$$d_n = n! \sum_{K=0}^{\infty} \frac{(-1)^K}{K!} \rightarrow (1)$$

For $n \geq 7$, the formula in (1) can be written as

$$d_n = \frac{n!}{e} = n! \times 0.3679$$

~~From this~~

1. Find the number of derangements of 1, 2, 3, 4

=> Let S be the set of all arrangements of 1, 2, 3, 4 then $|S| = N = S_0 = 4!$

$\frac{2 \times 3}{2 \times 1} = 6.$

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Let C_1, C_2, C_3, C_4 denotes the 1 is in its right place, 2 in the right place, 3 in the right place and 4 in the right place respectively. By the principle of inclusion and exclusion we have

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4$

Here $S_i = \sum N(C_i)$.

$N(C_1) = 3!$

$\therefore S_1 = {}^4C_1 \times 3!$

$N(C_2) = 2!$

$\therefore S_2 = {}^4C_2 \times 2!$

$N(C_3) = 1!$

$\therefore S_3 = {}^4C_3 \times 1! = 1$

$N(C_4) = 1!$

$\therefore S_4 = {}^4C_4 \times 0! = 1!$

$\therefore \bar{N} = 9.$

Note:-

2. In how many ways $1, 2, 3, \ldots, 10$ be arranged in a line so that no even integer is in its right place.

$\Rightarrow$ Let S be the set of all arrangements of the given digits $1, 2, 3, \ldots, 10$

$\therefore |S| = N = S_0 = 10!$

But a_i, $1 \le i \le 5$ denotes 2, 4, 6, 8, 10 in the right place respectively

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By the principle of inclusion and exclusion,

$S_i = \Sigma N(C_i)$

$N(C_1) = 9!$

$\therefore S_1 = {}^5C_1 \times 9!$

$S_2 = \Sigma N(C_i C_j)$

$N(C_1 C_2) = 8!$

$\therefore S_2 = {}^5C_2 \times 8!$

$S_3 = {}^5C_3 \times 7!$

$S_4 = {}^5C_4 \times 6!$

$S_5 = {}^5C_5 \times 5!$

$\bar{N} = S_0 - S_1 + S_2 - S_3 + S_4 - S_5$

$= 10! - \dots$

Note:- 2. The probability of derangement of 'n' objects is $\frac{d_n}{n!}$

3. A machine that inserts letters into envelops goes haywire and inserts letters randomly into envelops. What is the prob. that in a group of 100 letters,

i) No letter is put into the correct envelop

ii) Exactly one letter is put into the correct envelop

iii) Exactly 98 letters are put into the co—r—

iv) " 99 " " " "

v) All letters are put into the correct envelops.

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=> i) The probability of no letter being put in the correct envelop is $\frac{d_{100}}{100!}$ (∵ the no. of favourable cases (ie., derangements) is d_{100} & the total no. of favourable cases is 100!)

ii) When exactly one letter is put correctly; the no. of derangements for the remaining 99 letters is d_{99}. This can happen in ${}^{100}C_1$ ways. So the required probability is $\frac{{}^{100}C_1 \times d_{99}}{100!}$

iii) $\frac{{}^{100}C_{98} \times d_2}{100!}$; when exactly 98 letters are put into the right envelop, the no. of derangements for the remaining two is d_2. This can happen in ${}^{100}C_{98}$ ways $\Rightarrow {}^{100}C_{98}/100!$ (∵ $d_2 = 1$)

iv) $\frac{{}^{100}C_{99} \times d_1}{100!} = 0$; when exactly 99 letters are put into the correct envelop, it is not possible to misplace the remaining 1 letter. This is also put into the right. Thus the prob. is zero.

v) When all letters are put into the correct envelop, this can be happen only out of the 100! cases. Thus the required probability is $\frac{1}{100!}$

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Rook polynomial

A rook polynomial is a polynomial $r(c,x) = 1 + rx + r_2x^2 + \dots + r_nx^n$ where c is the chess board & r_n denotes the no. of ways in which n rooks (pawns) can be placed on a board such that no two rooks are in the same row or in the same column of the board.

1. Find the rook polynomials for the following chess boards.

i)

1	2
3	4

Here $r_0 = 1$

$r_1 = 4$

$r_2 = 2$ ($\because$ (1,4) and (2,3))

$r_3 = 0$.

$r(c,x) = 1 + 4x + 2x^2$

iii)

	1
	2
4	3

Here $r_0 = 1$

$r_1 = 4$

$r_2 = 2$ ($\because$ (1,4) and (2,4))

$\therefore r(c,x) = 1 + 4x + 2x^2$

iii)

1	2	3
4	X	X
X	5	6

Here $r_0 = 1$

$r_1 = 6$

$r_2 = (1,5)(1,6)(2,4)(2,6)(3,4)(3,5)(4,5)(4,6) = 8$

$r_3 = (2,4,6)$ ~~$(3,4,6)$~~ $(3,4,5) = 2$.

$\therefore r(c,x) = 1 + 6x + 8x^2 + 2x^3$

iv)

1	2	3
4		5

$r_1 = 5,\ r_0 = 1,\ r_2 = 4$

$r(C,x) = 1 + 5x + 4x^2$

v)

1	2	
		3
4	5	6

$r(C,x) = 1 + 6x + 8x^2 + 2x^3$

vi)

	1	2
	3	4
5	6	7

$r(C,x) = 1 + 7x + 10x^2 + 2x^3$

vii)

1	2	3
4		5
6	7	8

$r(C,x) = 1 + 8x + 14x^2 + 4x^3$

(2) Find the rook polynomial for 3×3 chess board

1	2	3
4	5	6
7	8	9

⇒ Here $r_0 = 1$

$r_1 = 9$

$r_2 = 18$

$r_3 = 6$

$r_4 = 0$

$\Rightarrow 1 + 9x + 18x^2 + 6x^3$

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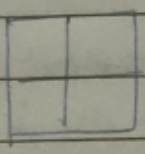
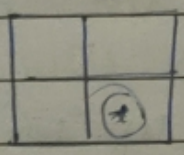
Expansion formula

In a given chess board C, suppose we choose a particular square and mark it as *. Let D be the board obtained from the board C by deleting the row and column containing the square marked by *. And let E be the board obtained from C by deleting only the square marked by *. Then the rook polynomial for the chess board C is $r(C,x) = x\, r(D,x) + r(E,x)$

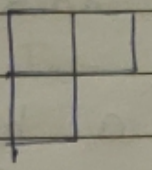
Ex:- $r(C,x) = ?x$

C: 

$r(C,x) = 1 + 4x + 2x^2$

C: 

D:

E: 

At D: $r_0 = 1,\ r_1 = 1$

$r(D,x) = 1 + x$

At E: $r_0 = 1,\ r_1 = 3,\ r_2 = 1$

$r(E,x) = 1 + 3x + x^2$

$r(C,x) = x\, r(D,x) + r(E,x)$

$= x[1+x] + 1 + 3x + x^2$

$= x + x^2 + 1 + 3x + x^2$

$= 1 + 4x + 2x^2$

Using the expansion formula find the rook polynomial for 3×3 chess board.

$\Rightarrow$ C:

1	2	3
4	5	6
7	8	9

D:

1	2
4	5

E:

1	2	3
4	5	6
7	8	

(1,5)(1,6)(1,8)
(2,4)(2,6)(2,7)
(3,4)(3,5)(3,7)
(5,7)(5,8)(4,8)(6,8)(6,7)

$r(D,x) = 1 + 4x + 2x^2$

$r(E,x) \Rightarrow r_0 = 1,\ r_1 = 8,\ r_2 = 14$

$r_3 = (1,6,8)(2,4,7)(3,4,8)$
$(3,5,7) = 4$

$\therefore r(E,x) = 1 + 8x + 14x^2 + 4x^3$

$r(C,x) = x\, r(D,x) + r(E,x)$
$= x[1 + 4x + 2x^2] + [1 + 8x + 14x^2 + 4x^2$
$= x + 4x^2 + 2x^3 + 1 + 8x + 14x^2 + 4x^2$
$= 1 + 9x + 18x^2 + 8x^3$

Suppose a board C is made up of two parts C_1 and C_2 where C_1 & C_2 have no ~~spare~~ squares in the same row or same column of the chess board C, then such parts of C are called ~~dis~~ design sub-boards of the chess board C. then the rook polynomial for the chess board C is $r(C,x) = r(C_1,x) \times r(C_2,x)$

Arrangements with forbidden positions

Suppose m objects are to be arranged in n places where $n \geq m$ suppose there are constraints under which some objects cannot

occupy certain places. Such places are called the forbidden positions for the set objects. The no of ways carrying out these toss is given by the following rule.

$\overline{N} = S_0 - S_1 + S_2 - S_3 + \cdots + (-1)^n S_n$

where $S_0 = n!$ and $S_k = (n-k)! r_k$ for $k = 1, 2, 3, \ldots n$

Here r_k is the coefficient of x^k in the rook polynomial of the board of 'm' rows and 'n' columns whose squares represent the forbidden places under specified conditions.

Problems. (For 7m)

1. The seating arrangements of 4 persons P_i $1 \le i \le 4$ at 5 different ~~places~~ tables T_j, $1 \le j \le 5$ with the following restrictions as

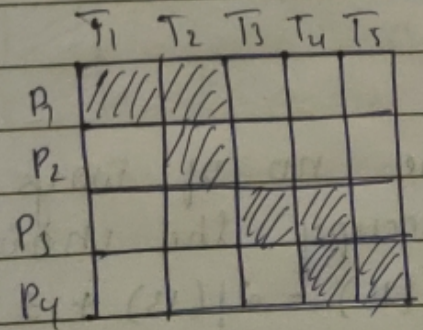
i) P_1 will not sit at T_1 or T_2

ii) P_2 will not sit at T_2

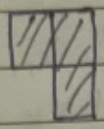
iii) P_3 —— " —— " —— T_3 or T_4

iv) P_4 —— " —— " —— T_4 or T_5

Find the no. of ways they can occupy the vacant chairs.

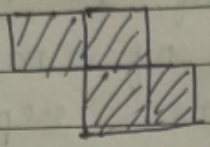


The shaded squares in the first row indicate the tables T_1 & T_2 are forbidden for 4 persons. Find the rook polynomial for the shaded squares. Here their exists 2 ~~disjoint~~ sub-boards

C_1: 

$r_0 = 1,\ r_1 = 3,\ r_2 = 1$

$\Rightarrow 1 + 3x + x^2$

C_2: 

$r_0 = 1,\ r_1 = 4,\ r_2 = 3$

$\Rightarrow 1 + 4x + 3x^2$

$$r(C, x) = r(C_1 x) \times r(C_2 x)$$
$$= (1 + 3x + x^2)(1 + 4x + 3x^2)$$
$$= 1 + 7x + 16x^2 + 13x^3 + 3x^4$$
$$\downarrow r_0 \quad \downarrow r_1 \quad \downarrow r_2 \quad \downarrow r_3 \quad \downarrow r_4$$

$S_0 = n! = 5!$

$S_k = (n-k)!\, r_k$

$S_1 = (5-1)!\, r_1$

$= 4!\,(7)$

$S_2 = (5-2)!\, r_2$

$= 3!\,(16)$

$S_3 = (5-3)!\, r_3$

$2!\,(13)$

$S_4 = (5-4)!\, r_4$

$= 3$

Consequently the no of ways in which four persons can occupy the chairs is

$5! - 4!\,(7) + 3!\,(16) - 2!\,(13) + 3$

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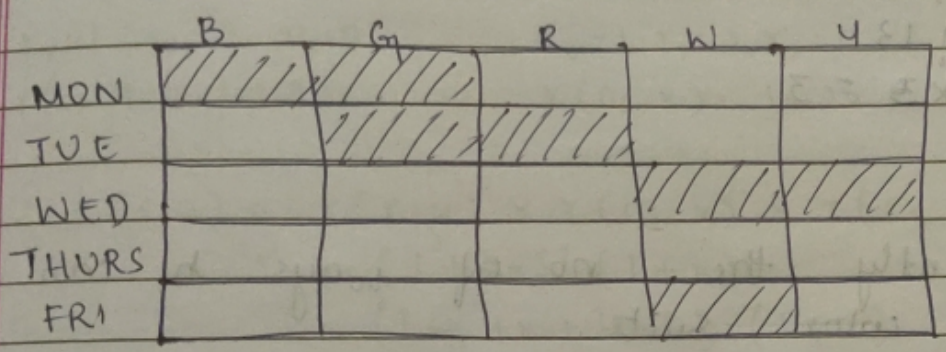
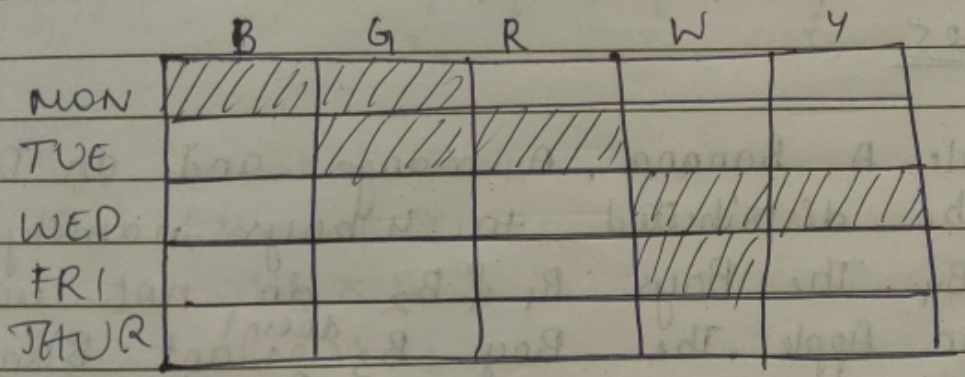
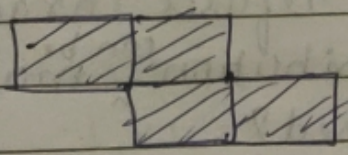
2 An executive attending a week long business seminar has 5 suits of different colors.
(4) On Mondays, he doesn't wear blue or green.
On Tuesdays, he doesn't wear red or green.
On Wednesdays, he doesn't wear yellow or white.
" Fridays " " " white. In how many ways can be dress with the above conditions satisfied.
Note :- 5 working days are taken as a week



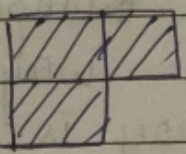
Find the rook polynomial for the shaded squares, then there exists two disjoint sub-boards



C_1 :



C_2 :



$r_0 = 1, r_1 = 4, r_2 = 3$

$1 + 4x + 3x^2$

$r_0 = 1, r_1 = 3, r_2 = 1$

$1 + 3x + x^2$

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$r[c,x] = r(c_1,x) \times r(c_2,x)$

$= (1+4x+3x^2) \times (1+3x+x^2)$

$= 1+7x+16x^2+13x^3+3x^4$

$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow \quad \downarrow$

$r_0 \quad r_1 \quad r_2 \quad r_3 \quad r_4 \quad ; n! = 5! = S_0$

$S_k = (n-k)! \, r_k$

$= (5-k)! \, r_k$

$S_1 = (4)! \, 7$

$S_2 = 3! \, 16$

$S_3 = 2! \, 13$

$S_4 = 1 \times 3 = 3$

$S_5 = 0.$

Consequently the no. of ways he can dress with 5 color suits

$\overline{N} = S_0 - S_1 + S_2 - S_3 + S_4$

$= \underline{\underline{25}}$

3. An Apple, A Banana, A mango and an Orange are to be distributed to 4 boys namely B_1, B_2, B_3, B_4. The Boys B_1 & B_2 do not wish to (5) have an Apple. The Boy B_3 doesn't want Banana and Mango and B_4 refuses orange. In how many ways the distribution can be made so that no boy is displeased?

$\Rightarrow$

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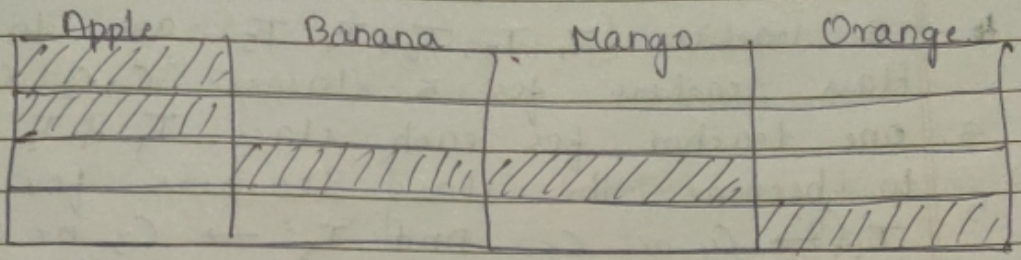
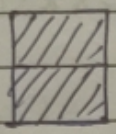
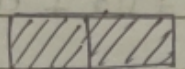
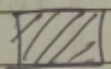


Find the rook polynomial for the shaded regions. Here there exists 3 disjoint sub-boards







$r_0 = 1, r_1 = 2, r_2 = 0$ $\quad$ $r_0 = 1, r_1 = 2, r_2 = 0$ $\quad$ $r_0 = 1, r_1 = 1$

$r(C_1, x) = 1 + 2x$ $\quad$ $r(C_2, x) = 1 + 2x$ $\quad$ $r(C_3, x) = 1 + x$

$$\therefore r(C, x) = r(C_1, x) \times r(C_2, x) \times r(C_3, x)$$
$$= (1 + 2x)^2 (1 + x)$$
$$= (1 + 4x + 4x^2)(1 + x)$$
$$= 1 + 5x + 8x^2 + 4x^3$$
$$\downarrow \quad \downarrow \quad \downarrow \quad \downarrow$$
$$r_0 \quad r_1 \quad r_2 \quad r_3$$

$n! = 4!$

$S_0 = 4!$

$S_k = (n-k)! \, r_k$

$S_1 = (4-1)! \, r_1$

$= 4! \times 5$

$S_2 = (4-2)! \, r_2$

$= 2! \times 8$

$S_3 = (4-3)! \, r_3$

$= 1 \times 4 = 4$

$N = S_0 + S_1 - S_2 + S_3 + S_4$

$= 6$

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4. 5 teachers T_1, T_2, T_3, T_4, T_5 are to be made class teachers for 5 classes C_1, C_2, C_3, C_4, C_5 one teacher for each class. T_1 & T_2 don't wish to become the class teachers for C_1 or C_2. $T_4 \xrightarrow{T_3} C_4$ or C_5. And $T_5 \rightarrow C_3$ or C_4 or C_5. In how many ways can the teachers be assigned the work without displeasing any teacher. $(1 + 11x + 40x^2 + 56x^3 + 28x^4 + 4x^5)$.

$\vec{N} = 8$
$n = 5$

Generating Functions.

Let $a_0, a_1, a_2, \ldots$ be a sequence of real numbers. The function $f(x) = a_0 + a_1 x + a_2 x^2 + \ldots = \sum_{i=0}^{\infty} a_i x^i$ is called the generating function for the given sequence.

1) Obtain the generating function for the sequence $\{1, 1, \ldots\}$

$f(x) = 1 + x + x^2 + x^3 \ldots = \frac{1}{1-x}$

2) Obtain the generating function for the seq. $\{1, -1, 1, -1, 1, -1 \ldots\}$

$\Rightarrow f(x) = 1 - x + x^2 - x^3 + x^4 = \frac{1}{1+x} \left[\because \left(\frac{1}{1-x}\right)^{-1} \right]$

3) Obtain the generating function for the sequence of natural numbers $= \{1, 2, 3, \ldots\}$

$\Rightarrow f(x) = 1 + 2x + 3x^2 + 4x^3 + \ldots$

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$= D[x + x^2 + x^3 + x^4 + \ldots]$ where $D = \frac{d}{dx}$

$= D(x)[1 + x + x^2 + x^3 + \ldots]$

$= D(x)\left[\frac{1}{1-x}\right] = \frac{1}{(1-x)^2}$

4) Obtain the generating function for the sequence $\{0, 1, 1, \ldots\}$

$\Rightarrow f(x) = 0 + x + x^2 + x^3 + \ldots$

$= x[1 + x + x^2 + \ldots]$

$= \frac{x}{1-x}$ //

5) $\{0, 1, -1, 1, -1, \ldots\}$

$\Rightarrow f(x) = 0 + x - x^2 + x^3 - x^4 + \ldots = \frac{x}{1+x}$ //

6) Find the generating function for the sequence $\{0^2, 1^2, 2^2, 3^2, \ldots\}$

$\Rightarrow f(x) = (0 + 1^2 x + 2^2 x^2 + 3^2 x^3 + \ldots)$

$= x[1^2 + 2^2 x + 3^2 x^2 + \ldots)$

$= x \cdot D[x + 2x^2 + 3x^2 + \ldots)$

$= x D(x)[1 + 2x + 3x^2 + \ldots]$

$= x D(x)\left[\frac{1}{1-x}\right]^2$

$= x \frac{d}{dx}\left(\frac{1}{(1-x^2)^2}\right) = \frac{x(1+x)}{(1+x)^3}$ //

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Find Prove that $1+x+x^2+\ldots+x^n = \frac{1-x^{n+1}}{1-x}$

$\Rightarrow \frac{1-x^{n+1}}{1-x} = \frac{(1-x^3)}{(1-x)} \quad 1-x^3-1x^3$

$= \frac{1}{1-x} - \frac{x^3}{1-x}$ =

Add and subtract $\rightarrow [x^{n+1} + x^{n+2} + \ldots]$

1. Find the sequence generated by the function $\frac{x}{(1-x)^2}$

$\Rightarrow f(x) = \frac{x}{(1-x)^2}$

$= x[1+2x+3x^2+4x^3+\ldots]$

$= 0+x+2x^2+3x^3$

$= \{0, 1, 2, 3, 4, \ldots\}$

2. $\frac{1}{1-x} + 3x$ $\quad [1+x+x^2+\ldots] + 3x$

$\Rightarrow \{1, 4, 1, 1, \ldots\}$

3. Find the Binomial coefficient if
i) n is not necessarily an integer (i.e., $n \in \mathbb{R}$)
ii) n is a negative integer.

$(-1)^r$

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i) ${}^nC_r = \frac{n!}{(n-r)!\,r!}$ $n, r \in z^+$ & $n \geq r \geq 0$

$= \frac{n(n-1)(n-2)\ldots[n-(r-1)](n-r)!}{(n-r)!\,r!}$

${}^nC_r = \frac{n(n-1)(n-2)\ldots[n-(r-1)]!}{r!}$

ii) ${}^nC_r = \frac{n(n-1)(n-2)\ldots[n-(r-1)]}{r!}$

replace n by $-n$, we get

${}^{-n}C_r = \frac{(-n)(-n-1)(-n-2)\ldots[-n-(r-1)]}{r!}$

$= \frac{(-1)[n+(r-1)]\ldots(-1)(n+2)(-1)(n+1)(-1)n}{r!}$

$= \frac{(-1)^r[n+(r-1)]\ldots(n+2)(n+1)n \times (n-1)!}{r!\,(n-1)!}$

$$\boxed{{}^{-n}C_r = \frac{(-1)^r[n+(r-1)]!}{r!\,(n-1)!}}$$ Hence

$= (-1)^r\ {}^{n+r-1}C_r$

$= (-1)^r\ c(n+r-1, r)$

$=$

4. Find the coefficient of x^5 in $[1-2x]^{-7}$

$\Rightarrow f(x) = (1-2x)^{-7} = \sum_{r=0}^{\infty} {}^{-7}C_r (-2x)^r$

$= \sum_{r=0}^{\infty} (-1)^r \; {}^{7+r-1}C_r \; (-1)^r (2x)^r$

$= \sum_{r=0}^{\infty} {}^{6+r}C_r \; (2x)^r$

For $r=5$ ${}^{6+5}C_5 (2x)^5 = {}^{11}C_5 \times 2^5 \times x^5$

$\therefore$ The coeff of x^5 is ${}^{11}C_5 \times 2^5$

5. Determine the coefficient of x^{15} in $f(x) = (x^2 + x^3 + x^4 + \dots)^4$

$\Rightarrow f(x) = (x^2 + x^3 + x^4 + \dots]^4$

$= [x^2 [1 + x + x^2 + \dots)]^4$

$= \left[x^2 \times \frac{1}{1-x} \right]^4$

$= \frac{x^8}{(1-x)^4}$

$\therefore f(x) = x^8 (1-x)^{-4}$

$f(x) = x^8 \sum_{r=0}^{\infty} {}^{-4}C_r (-x)^r$

$= x^8 \sum_{r=0}^{\infty} (-1)^r \; {}^{4+r-1}C_r \; (-1)^r x^r$

$= x^8 \sum_{r=0}^{\infty} {}^{3+r}C_r \; x^r$

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$$= \sum_{r=0}^{\infty} {}^{3+r}C_r \, x^{r+8}$$

$r = 7$, $\quad {}^{3+7}C_7 \, x^{7+8} = {}^{10}C_7 \, x^{15}$

Ans $= {}^{10}C_7$

6. x^{50} in $[x^7 + x^8 + x^9 + x^{10} + \ldots]^6$.

$\Rightarrow$ Let $f(x) = (x^7 + x^8 + x^9 + x^{10} + \ldots]^6$

$= [x^7(1 + x^7 + x^8 + x^9 + \ldots)]^6$

$= \left[x^7 \times \left(\frac{1}{1-x}\right)\right]^6$

$= \frac{x^{42}}{(1-x)^6}$

$= x^{42}[1-x]^{-6}$

$f(x) = x^{42} \sum_{r=0}^{\infty} {}^{-6}C_r \, (-x)^r$

$= x^{42} \sum_{r=0}^{\infty} (-1)^r \, {}^{6+r-1}C_r \, (-1)^r \, x^r$

$= \sum_{r=0}^{\infty} {}^{5+r}C_r \, x^{r+42}$.

For $r = 8$, we have

$5 + {}^{8}C_8 \, x^{8+42} = {}^{13}C_8 \, x^{50}$.

$\Rightarrow {}^{13}C_8$

7. x^{20} in $(x^2 + x^3 + x^4 + x^5 + x^6)^5$

(1)

$\Rightarrow$ Let $f(x) = (x^2 + x^3 + x^4 + x^5 + x^6)^5$

$= [x^2(1 + x + x^2 + x^3 + x^4)]^5$

$= \left[x^2\left(\frac{1 - x^5}{1 - x}\right)\right]^5$

$= \frac{x^{10}[1 - x^5]^5}{[1 - x]^5}$

$f(x) = x^{10} \sum_{r=0}^{\infty} {}^{-5}C_r (-x)^r \cdot {}^{5}C_r (-x^5)^5 \sum_{k=0}^{\infty} {}^{-5}C_k (-x)^k$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-1)^r (-1)^{5+r} \sum_{k=0}^{\infty} {}^{-5}C_k (-1)^k x^{10+r+k}$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-1)^{5-r} x^{25-5r} \sum_{k=0}^{\infty} (-1)^k {}^{5-k-1}C_k (-1)^k x^k$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-1)^{5-r} x^{25-5r} \sum_{k=0}^{\infty} {}^{4+k}C_k x^k$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-x^5)^r \sum_{k=0}^{\infty} {}^{-5}C_k (-x)^k$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-1)^r x^{5r} \sum_{k=0}^{\infty} (-1)^k {}^{5+k-1}C_k (-1)^k x^k$

$= x^{10} \sum_{r=0}^{5} {}^{5}C_r (-1)^r x^{5r} \sum_{k=0}^{\infty} {}^{4+k}C_k x^k$

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case (i): $r=2$ and $k=0$ we have

$x^{10}\ {}^5C_2\ (-1)^2\ x^{10} = {}^5C_2\ x^{20}$.

case (ii) $r=0$ and $k=10$, we have

$x^{10} \times {}^{4+10}C_{10}\ x^{10} = {}^{14}C_{10}\ x^{20}$.

case (iii) $r=1$ and $k=5$ we have

$x^{10} \times {}^5C_2(-1)^1\ x^5 \times {}^{4+5}C_5\ x^5$

$= {}^5C_1 \times {}^9C_5 \times x^{20}$.

$= {}^5C_2 + {}^{14}C_{10} - ({}^5C_1 \times {}^9C_5)$

8. Find the coefficient of x^{24} in the expansion of $[x^3 + x^4 + \dots + x^8]^4$

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9. Find the no of ways to select 3 books from 6 distinct books

=> ~~The generating function~~
For each of the books there are 2 possibilities that a book is selected or not and which can be represented as a power series of the form $x^0 + x^1$. ∴ The generating function for the given problem is $(x^0+x^1)^6 = [1+x]^6$

$$= \sum_{r=0}^{6} {}^6C_r\, x^r$$

$r = 3 \Rightarrow {}^6C_3\, x^3$

∴ ${}^6C_3$ ways to select three books from six distinct books

10 In how many ways can 12 marbles be distributed among 3 students A, B, C so that A gets atleast 4, B and C gets atleast 2 so that C gets not more than 5 marbles.

(8)

=> $A: x^4 + x^5 + x^6 + \dots = x^4[1 + x + x^2 + \dots] = \frac{x^2}{1-x}$

$B: x^2 + x^3 + x^4 + \dots = \frac{x^2}{1-x}$

$C: x^2 + x^3 + x^4 + x^5 = x^2[1 + x + x^2 + x^3] = x^2\left[\frac{1-x^4}{1-x}\right]$

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∴ The generating function for the given prob is

$$f(x) = \frac{x^4}{1-x} \times \frac{x^2}{1-x} \times x^2 \left[\frac{1-x^4}{1-x}\right]$$

$$= \frac{x^8(1-x^4)}{(1-x)^3}$$

$$f(x) = (x^8 - x^{12})(1-x)^{-3}$$

$$= x^8(1-x)^{-3} - x^{12}(1-x)^{-3}$$

$$= x^8 \sum_{r=0}^{\infty} {}^{-3}C_r (-x)^r - x^{12} \sum_{k=0}^{\infty} {}^{-3}C_k (-x)^k$$

$$f(x) = x^8 \sum_{r=0}^{\infty} (-1)^r \, {}^{3+r-1}C_r (-1)^r x^r - x^{12} \sum_{k=0}^{\infty} (-1)^k \, {}^{3+k-1}C_k (-1)^k x^k$$

$$f(x) = x^8 \sum_{r=0}^{\infty} {}^{2+r}C_r \, x^r - x^{12} \sum_{k=0}^{\infty} {}^{2+k}C_k \, x^k.$$

For $r=4$ and $k=0$, we have

$$x^8 \; {}^{2+4}C_4 \, x^4 - x^{12} = \left[{}^6C_4 - 1\right] x^{12}$$

∴ $({}^6C_4 - 1)$ ways to distribute the 18 marbles to 3 students of desired type

11. In how many ways can a police captain distribute 24 Rifle shells to 4 police officers so that each officer gets atleast 3 shells but not more than 8

$\Rightarrow$ The choices for the number of shells that each officer receives is given by $(x^3+x^4+\dots+x^8)$ There are 4 officers so the resulting generating function $f(x) = (x^3+x^4+\dots+x^8)^4$

$$= [x^3(1+x+\dots x^5)]^4$$

$$= \left[x^3\left(\frac{1-x^6}{1-x}\right)\right]^4$$

$$= \frac{x^{12}(1-x^6)^4}{(1-x)^4}$$

$$f(x) = x^{12} \sum_{r=0}^{4} {}^4C_r (-x^6)^r \sum_{k=0}^{\infty} {}^{-4}C_k (-x)^k$$

$$= x^{12} \sum_{r=0}^{4} {}^4C_r (-1)^r x^{6r} \sum_{k=0}^{\infty} (-1)^k {}^{4+k-1}C_k (-1)^k x^k$$

$$= x^{12} \sum_{r=0}^{\infty} {}^4C_r (-1)^r x^{6r} \sum_{k=0}^{9} {}^{3+k}C_k x^k$$

case (1) = for $r=2$ and $k=0$ we have

$$= x^{12} {}^4C_2 (-1)^2 x^{12} \times 1 = {}^4C_2 \times x^{24}$$

case (2) = for $r=0$ and $k=12$ we have

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$x^{12} \times 1 \times {}^{3+12}C_{12}\, x^{12} = {}^{15}C_{12}\, x^{24}$.

case (3) = for $r = 1$ and $k = 6$ we have

$x^{12}\ {}^{4}C_1 (-1)^1 x^6 \times {}^{3+6}C_6\, x^6 = -{}^4C_1 \times {}^9C_6\, x^{24}$

$\therefore$ The coefficient of x^{24} is

$= {}^4C_2 + {}^{15}C_{12} - {}^4C_1 \times {}^9C_6$.

12. (10) In how many can 3000 identical envelops be divided into ~~tax~~ packs of 25 among 4 groups so that each group gets atleast 150 envelops but not more than 1000 envelops

$\Rightarrow$ The no. of packets $= \frac{3000}{25} = 120$

Each group gets atleast $\frac{150}{25}$ envelops = 6 packets

but not more than $\frac{1000}{25}$ envelops = 40 packets

$\therefore$ The generating function for the given prob. is $f(x) = [x^6 + x^7 + \ldots + x^{40}]^4$

$f(x) = [x^6(1 + x + x^2 + \ldots + x^{34})]^4$

$= \left(x^6 \left[\frac{1 - x^{35}}{1 - x}\right]\right)^4$

$= x^{24}\left(\frac{(1 - x^{35})}{(1 - x)}\right)^4 \Rightarrow x^{24}(1 - x^{35})^4 (1 - x)^{-4}$

$$= x^{24} \sum_{r=0}^{4} {}^4C_r (-x^{35})^r \sum_{k=0}^{\infty} {}^{-4}C_k (-x)^k$$

$$= x^{24} \sum_{r=0}^{4} {}^4C_r (-1)^r x^{35r} \sum_{k=0}^{\infty} {}^{3+k}C_k x^k$$

case1 = for $r=0$ & $k=96$ we have

$$x^{24} \times 1 \times {}^{3+96}C_{96} x^{96} = {}^{99}C_{96} x^{120}$$

case2 = for $r=1$ & $k=61$ we have

$$x^{24} \times {}^4C_1 (-1) x^{35} \times {}^{3+61}C_{61} x^{61} = -{}^4C_2 \times {}^{64}C_{61} x^{120}.$$

case3 = for $r=2$ & $k=26$ we have.

$$x^{24} \times {}^4C_2 (-1)^2 x^{70} \times {}^{3+26}C_{26} x^{26} = {}^4C_2 \times {}^{29}C_{26} x^{120}$$

∴ The coefficients of x^{120} is

$${}^{99}C_{96} + -[{}^4C_2 \times {}^{64}C_{61}] + [{}^4C_2 \times {}^{29}C_{26}]$$

13. In how many ways can 2 dozen identical Robots be assigned to 4 assembly lines with
   i) atleast 3 robots assigned to each line
   ii) atleast 3 but not more than 9 robots assigned to each line.

i) Given that $f(x) = [x^3 + x^4 + x^5 + \dots]^4$

$$= [x^3(1 + x + x^2 + \dots)]^4$$

$$= \left[x^3 \left(\frac{1}{1-x}\right)\right]^4$$

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$$= \frac{x^{12}}{(1-x)^4}$$

$$f(x) = x^{12}(1-x)^{-4}$$

$$= x^{12} \sum_{r=0}^{\infty} {}^{-4}C_r (-x)^r$$

$$= x^{12} \sum_{r=0}^{\infty} (-1)^r \, {}^{4+r-1}C_r \, (-1)^r x^r$$

$$= x^{12} \sum_{r=0}^{\infty} {}^{3+r}C_r \, x^r$$

for $r = 12$, we have

$$x^{12} \; {}^{3+12}C_{12} \; x^{12} = {}^{15}C_{12} \, x^{24}.$$

∴ The arrangement of 24 robots of desired type = ${}^{15}C_{12}$.

ii) Given that

$$f(x) = [x^3 + x^4 + x^5 + \cdots + x^9]^4$$

$$= [x^3(1 + x + x^2 + \cdots + x^6)]^4$$

$$= \left(x^3 \left[\frac{1-x^7}{1-x}\right]\right)^4$$

$$= \frac{x^{12}(1-x^7)^4}{(1-x)^4}$$

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$$f(x) = x^{12}(1-x^7)^4(1-x)^{-4}$$

$$= x^{12} \sum_{r=0}^{4} {}^4C_r (-x^7)^r \sum_{k=0}^{\infty} {}^{-4}C_k (-x)^k$$

$$= x^{12} \sum_{r=0}^{4} (-1)^r {}^4C_r x^{7r} \sum_{k=0}^{\infty} {}^{3+k}C_k x^k$$

case 1 = For $r=0$ & $k=12$

$$x^{12} \times 1 \times {}^{3+12}C_{12} \times x^{12} = {}^{15}C_{12} x^{24}$$

case 2 = For $r=1$ & $k=5$

$$x^{12} (-1) {}^4C_1 x^7 \times {}^{3+5}C_5 x^5$$

$$-{}^4C_1 \times {}^8C_5 x^{24}.$$

$\therefore$ Ans :- ${}^{15}C_{12} - ({}^4C_1 \times {}^8C_5)$

Note :- $e^x = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\ldots$

$$e^{-x} = 1 - \frac{x}{1!} + \frac{x^2}{2!} - \frac{x^3}{3!} + \ldots\ldots$$

$$\frac{1}{2}[e^x + e^{-x}] = 1 + \frac{x^2}{2!} + \frac{x^4}{4!} + \frac{x^6}{6!} + \ldots$$

$$\frac{1}{2}[e^x - e^{-x}] = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots$$

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Exponential generating function.

For a sequence $a_0, a_1, a_2 \ldots$ of real numbers. $f(x) = a_0 + a_1 \frac{x}{1!} + a_2 \frac{x^2}{2!} + \ldots$

$= \sum_{r=0}^{\infty} a_r \frac{x^r}{r!}$ is called the exponential generating function for the given sequence

Note:- If $a_0, a_1, a_2 \ldots$ be a sequence of real numbers for exponential generating function then $a_0, a_1, \frac{a_2}{2!}, \frac{a_3}{3!} \ldots$ be a sequence of real numbers for ordinary generating function

Problems

1. Find the exponential generating function for the sequence 1, 1, 1, 1, ...

$\Rightarrow$ we have, $f(x) = a_0 + a_1 x + a_2 \frac{x^2}{2!} + a_3 \frac{x^3}{3!} + \ldots$

$\Rightarrow f(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = e^x$.

2. 1, -1, 1, -1, ...

$\Rightarrow f(x) = 1 - x + \frac{x^2}{2!} - \frac{x^3}{3!} = e^{-x}$

$e^{ax} = 1 + \frac{ax}{1!} + \frac{}{2!}$

3. Find the exponential generating function for the sequence $1, 2, 2^2, 2^3, 2^4, \ldots$

$$\Rightarrow f(x) = a_0 + a_1 x + \frac{a_2 x^2}{2!} + \frac{a_3 x^3}{3!} + \frac{a_4 x^4}{4!} + \ldots$$

$$f(x) = 1 + 2x + \frac{2^2 x^2}{2!} + \frac{2^3 x^3}{3!} + \frac{2^4 x^4}{4!} + \ldots$$

$$= 1 + 2x + \frac{(2x)^2}{2!} + \frac{(2x)^3}{3!} + \frac{(2x)^4}{4!} + \ldots$$

$$= e^{2x}.$$

4. Determine the sequence generated by ~~the~~ each of the following exponential generating functions.

1) $f(x) = 5e^{5x}$ 2) $f(x) = e^{3x} - 28x^3 - 6x^2 + 9x$

3) $f(x) = \frac{1}{1-x}$ 4) $f(x) = \frac{3}{1-2x} + e^x$

1) Given that $f(x) = 5e^{5x} = 5 \sum_{r=0}^{\infty} \frac{(5x)^r}{r!}$

$$= 5\left[1 + \frac{5x}{1!} + \frac{(5x)^2}{2!} + \frac{(5x)^3}{3!} + \ldots\right]$$

$$= 5 + \frac{5^2 x}{1!} + \frac{5^3 x^2}{2!} + \frac{5^4 x^3}{3!} = 5, 5^2, 5^3, 5^4, \ldots$$

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2) $f(x) = e^{3x} - 28x^3 - 6x^2 + 9x$

$= \left[1 + \frac{3x}{1!} + \frac{(3x)^2}{2!} + \frac{(3x)^3}{3!} + \frac{(3x)^4}{4!} + \ldots\right]$

$- 28x^3 - 6x^2 + 9x$

$= \left[1 + 3x + \frac{3^2x^2}{2!} + \frac{3^3x^3}{3!} + \frac{3^4x^4}{4!} + \ldots\right] - \frac{(28 \times 3!)x^3}{3!}$

$- \frac{(6 \times 2!)x^2}{2!} + 9x$

$= 1 + 12x + \frac{(-3)x^2}{2!} + \frac{(-141)x^3}{3!} + 3^4x^4 + \ldots$

$\therefore = 1, 12, -3, -141, 3^4, 3^5, \ldots$

3) $f(x) = \frac{1}{1-x} = 1 + x + x^2 + x^3 + \ldots$

$= 1 + x + \frac{2!x^2}{2!} + \frac{3!x^3}{3!} + \ldots$

Given $f(x)$ is an G.F for the sequence

$1, 1, 2!, 3!, \ldots$

4) $4, 7, 25, 145, \ldots$

Ans:

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5. (12) Using the exponential generating function find the number of ways in which 4 of the letters in the word ENGINE

=> We have E → 2, N → 2, G = I = 1
In an arrangements of these letters the letter E can appear 0, 1, or 2 times. The exponential generating function corresponding to letter E is $f_1(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$ =>

Similarly the E G E corresponding to letter N is $f_2(x) = 1 + \frac{x}{1!} + \frac{x^2}{2!}$

Since the letter G can appear 0, or 1 time in an arrangement then the corresponding exponential generating function is $f_3(x) = 1 + \frac{x}{1!}$

Also the letter I can apperas 0 or 1 time in an arrangement. Hence $f_4(x) = 1 + \frac{x}{1!}$

Then $f(x) = f_1(x)\, f_2(x)\, f_3(x)\, f_4(x)$

$$= \left(1 + x + \frac{x^2}{2!}\right)^2 (1+x)^2$$

$$= \left(1 + 2x + 2x^3 + x^3 + \frac{x^4}{4!}\right)\left(1 + 2x + 2x^2 + 0x^3 + 0x^4\right)$$

We have, $x^4\left[(1\times 0) + (2\times 0) + (1\times 2) + (1/4 \times 1)\right]$

$x^4\left[2 + 2 + 1/4\right] = x^4 \times 4! \left(\frac{11}{4}\right)$

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∴ The coefficient of $\frac{x^4}{4!} = 4! \times \frac{17}{4} = \underline{\underline{102}}$

6. Using the E.G.F find the no. of ways in which 4 of the letters in the word HAWAII

⇒ Ans :- 102

H A W A I I

H = W = 1, A → 2, I → 2

$f_1(x) = 1 + x + \frac{x^2}{2!}$

$f_2(x) = 1 + x + \frac{x^2}{2!}$

$f_3(x) = 1 + x = f_4(x)$.

$f(x) = f_1(x) + f_2(x) + f_3(x) + f_4(x)$.

$= \left(1 + x + \frac{x^2}{2!}\right)^2 (1+x)^2$

$= \frac{(1 + 2x + 2x^3 + x^3 + x^4)}{4!}(1 + 2x + x^2 + 0x^3 + 0x^4)$

$= x^4 [(1 \times 0) + (2 \times 0) + (1 \times 2) + (1/4 \times 1)$

$= x^4 (2 + 2 + 1/4) = x^4 \times 4! (17/4)$

∴ The coefficient of $\frac{x^4}{4!} = 4! \times \frac{17}{4} = \underline{\underline{102}}$

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7. Find the E.G.F for the no. of ways to arrange n letters where $n \geq 0$ selected from each of the following boards

i) MISSISSIPPI

ii) ISOMORPHISM

i) $M \to 1,\ I \to 4,\ S \to 4,\ P \to 2$

$$f(x) = \left[1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \frac{x^4}{4!}\right]^2 \left[1 + x + \frac{x^2}{2!}\right]\left[1 + x\right]$$

ii) $I \to 2,\ S \to 2,\ O \to 2,\ M \to 2,\ R \to 1,\ P \to 1,\ H \to 1,$

$$f(x) = \left(1 + x + \frac{x^2}{2!}\right)^4 (1 + x)^3$$

8. A ship carries 48 flags, 12 each of the colors Red, White, Blue and Black. 12 of these flags are placed on a vertical pole in order to communicate a signal to other ships

i) How many of these signals use an even number of Blue flags and an odd number of black flags.

There is no restriction on rest of the colors

ii) How many of the signals have atleast 3 white flags or no white flags at all.

=> i) $$f_1(x) = x^0 + \frac{x^2}{2!} + \frac{x^4}{4!} + \ldots = \frac{1}{2}\left[e^x + e^{-x}\right]$$

$$f_2(x) = \frac{x}{1!} + \frac{x^3}{3!} + \frac{x^5}{5!} + \ldots = \frac{1}{2}\left[e^x - e^{-x}\right]$$

$$f_3(x) = 1 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots = e^x = f_4(x)$$

$f(x) = f_1(x)\, f_2(x)\, f_3(x)\, f_4(x)$

$f(x) = \frac{1}{4}\left(e^{2x} - e^{-2x}\right)\left(e^{2x}\right)$

$= \frac{1}{4}\left(e^{4x} - 1\right)$

$= \frac{1}{4} e^{4x} - \frac{1}{4}$

$= \frac{1}{4} \sum_{r=0}^{\infty} \frac{(e^{4x})^r}{r!} - \frac{1}{4}$.

ii) Corresponding to the blue flags which are to be even in number, then we have corresponding to ~~be~~ black flags which are to be odd in number, then we have Since there are no restrictions on red and white flags, corresponding to these we have

The coefficient of $\frac{x^{12}}{12!} = \frac{1}{4} \times 4^{12} = 4^{11}$

iii) In this case we have $f(x) = \left[x^0 + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots\right]^1 \cdot$

$\left[x^0 + x + \frac{x^2}{2!} + \frac{x^3}{3!} + \ldots\right]^3$

$= \left[1 + \frac{x^3}{3!} + \frac{x^4}{4!} + \ldots + x + \frac{x^2}{2!} - x - \frac{x^2}{2!}\right](e^x)^3$

$= \left[e^x - x - \frac{x^2}{2!}\right] e^{3x}$,

$f(x) = e^{4x} - x.e^{3x} - \frac{x^2}{2!} e^{3x}$.

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$$f(x) = \sum_{r=0}^{\infty} \frac{(4x)^r}{r!} - x \sum_{k=0}^{\infty} \frac{(3x)^k}{k!} - \frac{x^2}{2} \sum_{l=0}^{\infty} \frac{(3x)^l}{l!}$$

For $r=12$, $k=11$, $l=10$, we have

$$= \left(\frac{4^{12} \cdot x^{12}}{12!}\right) - x\left(\frac{3^{11} x^{11}}{11!}\right) - \frac{x^2}{2}\left(\frac{3^{10} x^{10}}{10!}\right)$$

$$= \left(\frac{4^{12} \times x^{12}}{12!}\right) - \left(\frac{3^{11} x^{12}}{12 \cdot 11}\right) - \frac{1}{2}\left(3^{10} \times \frac{x^{12}}{10!}\right)$$

$$= \left(\frac{4^{12} \times x^{12}}{12!}\right) - \left(\frac{3^{11} \times 12 \times x^{12}}{12!}\right) - \frac{1}{2}\left(\frac{3^{10} \times 12 \times 11 \times x^{12}}{12!}\right)$$

$$= \left[4^{12} - (3^{11} \times 12) - (3^{10} \times 6 \times 11)\right] \frac{x^{12}}{12!}$$

The cost of $\frac{x^{12}}{12!} = 4^{12} - (3^{11} \times 12) - (3^{10} \times 66)$

Partitions of Integers

Partitions of a positive integer n is a collection of positive integer parts with sum equal to n. The no. of partitions of an integer n is denoted by $p(n)$.

For eg:-

1	1	$p(1) = 1$
2	2, 1+1	$p(2) = 2$
3	3, 2+1, 1+1+1	$p(3) = 3$
4	4, 2+2, 2+1+1, 1+1+1+1	$p(4) = 5$
5		$p(5) = 7$.

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We should like to obtain $p(n)$ for a given n without having to list all partitions. We need a tool to keep track of the numbers of 1s, 2s, 3s, ... n's that are used as sum ends for n. If $n \in Z^+$ the no of 1s we can use is 0 or 1 or 2 or ... times. The power series representation of the above case is $x^1 : (x^1)^0 + (x^1)^1 + (x^1)^2 + \ldots = \frac{1}{1-x} (1 + x + x^2 \ldots)$

The no. of 2s we can use is 0 or 1 or 2 or ... times. The power series representation of the above case can be written as

$x^1 : (x^2)^0 + (x^2)^1 + (x^2)^2 + \ldots = \frac{1}{1-x^2}$

The no. of 3s we can use is 0 or 1 or 2 or ... times. The power series representation of the above case can be written as

$x^3 : (x^3)^0 + (x^3)^1 + (x^3)^2 + \ldots = \frac{1}{1-x^3}$

.....

The no. of n's we can use is 0 or 1 or 2 or ... times. The power series is

$x^n : (x^n)^0 + (x^n)^1 + (x^n)^2 + \ldots = \frac{1}{1-x^n}$

The generating function for the no. of partitions of any positive integer n is

$p(n) = f(x) = \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^3} \times \ldots \times \frac{1}{1-x^n} \ldots$

This we can write it as $\prod_{k=1}^{\infty} \frac{1}{1-x^k}$

$$\cdot \prod_{k=1}^{\infty} (1-x^k)^{-1}$$

$$= \prod_{k=1}^{\infty} \sum_{r=0}^{\infty} x^{rk}$$

$(1-x)^{-1} \times (1-x^2)^{-1}$

$\therefore (1+x+x^2+\dots) \times (1+x^2+\dots)$

Note: There is no simple formula for $p(n)$. further the coefficient of x^n is $p(n)$

1. In order to determine $p(3)$ we have to find the coefficient of x^3 $p(x) = f(x) = \prod_{k=1}^{\infty} \sum_{r=0}^{\infty} x^{rk}$

$$f(x) = \prod_{k=1}^{\infty} \left(\frac{1}{1-x^2}\right) = \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^3} \times \dots$$

$$= (1+x+x^2+x^3+\dots)(1+x^2+(x^2)^2+\dots)(1+x^3+(x^3)^2+\dots)$$

$$= (1+x+x^2+x^3)(1+x^2)(1+x^3)$$

$3 \to 2+1, 3, 1+1+1$

$$= (1+x+x^2+x^3)(1+x^2+x^3)$$

$\therefore$ The coeff of x^3 is $x^3[1+1+1] = 3x^3$

$p(3) = 3$

2. For $n=6$ we have the following partitions

$\Rightarrow$ $6 \to 6, 5+1, 3+3, 4+2, 3+2+1, 3+1+1+1, 4+1+1$

$2+2+2, 2+2+1+1, 2+1+1+1+1, 1+1+1+1+1+1$

$\therefore p(6) = 11$

$P_d(n) = P_o(n)$

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Out of 11 partitions of an int. 6 the following partitions represent the no. of partitions of 6 into distinct parts or distinct sumends

6. 5+1. 3+2+1, 4+2

The no. of partitions of a +ve int n into distinct parts / distinct sumends can be denoted by $P_d(n)$ ∴ $P_d(6) = 4$

Out of 11 partitions of an integer 6 the following partition represent the number of partitions of 6 into odd sumends. and it is denoted by $P_o(6)$

$P_o(6) = 4$

5+1, 3+3, 3+1+1, 1+1+1+1+1+1

3 Find the generating function for $P_d(n)$ the no. of partitions of n into distinct sumends.

⇒ For any +ve integer k, the power series is $(x^k)^0 + (x^k)^1$ ∴ $f(x) = [(x^1)^0 + (x^1)^1][(x^2)^0 + (x^2)^1][(x^3)^0 + (x^3)^1]$

$= [1+x][1+x^2][1+x^3]\ldots$

$\therefore \prod_{k=1}^{\infty} (1+x^k) = P_d(x)$.

Here $P_d(n)$ is the coeff of x^n in $(1+x)(1+x^2)\ldots(1+x^n)$

4. Find the generating function for $P_o(n)$. The no. of partitions of x into odd sumends. Hence show that $P_o(n) = P_d(n)$

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$\Rightarrow$ Let $P_o(x)$ be the generating function for the no of partitions of an integer into odd summ

$1 \rightarrow (x^1)^0 + (x^1)^1 + (x^1)^2 + \cdots = \frac{1}{1-x}$

$3 \rightarrow (x^3)^0 + (x^3)^1 + (x^3)^2 + \cdots = \frac{1}{1-x^3}$

$5 \rightarrow (x^5)^0 + (x^5)^1 + (x^5)^2 + \cdots = \frac{1}{1-x^5}$

$$P_o(x) = \frac{1}{1-x} \times \frac{1}{1-x^3} \times \frac{1}{1-x^5} \cdots = \prod_{k=1}^{\infty} \frac{1}{1-x^{2k-2}} \rightarrow (1)$$

Here $P_o(n)$ is the coeff of x^n in (1) Now we have to s.t. $P_o(n) = P_d(n)$.

$$P_d(x) = (1+x)(1+x^2)(1+x^2) \cdots = \prod_{k=1}^{\infty} (1+x^k)$$

$$1+x = \frac{1-x^2}{1-x}; \quad 1+x^2 = \frac{1-x^4}{1-x^2}; \quad 1+x^3 = \frac{1-x^6}{1-x^3}$$

$$1+x^4 = \frac{1-x^8}{1-x^4}; \quad 1+x^5 = \frac{1-x^{10}}{1-x^5}$$

$$P_d(x) = (1+x)(1+x^2)(1+x3)(1+x^4)(1+x^5 \cdots$$

$$= \frac{1-x^2}{1-x} \times \frac{1-x^4}{1-x^2} \times \frac{1-x^6}{1-x^3} \times \frac{1-x^8}{1-x^4} \times \frac{1-x^{10}}{1-x^5}$$

$$= \frac{1}{1-x} \times \frac{1}{1-x^3} \times \frac{1}{1-x^5} \times \cdots = P_o(x)$$

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From the equality of the generating function $P_o(x) = P_d(x)$ we have $P_o(n) = P_d(n) \quad \forall \quad n \geq 0$.

5. Find the generating function for the no. of partitions where each summands must appear an even no. of times and also find the generating function where each summand must be odd and must appear and odd no. of times

⇒ Let $f(x)$ be the generating function for the no. of partitions of an integer n where each summand must appear an even no. of times.

$$1 \rightarrow (x^1)^0 + (x^1)^2 + (x^1)^4 + \ldots = \frac{1}{1-x^2}$$

$$2 \rightarrow (x^2)^0 + (x^2)^2 + (x^2)^4 + \ldots = \frac{1}{1-x^4}$$

$$3 \rightarrow (x^3)^0 + (x^3)^2 + (x^3)^4 + \ldots = \frac{1}{1-x^6}$$

$$f(x) = \frac{1}{1-x^2} \times \frac{1}{1-x^4} \times \frac{1}{1-x^6} \times \ldots = \prod_{k=1}^{\infty} \frac{1}{1-x^{2k}}$$

Let $g(x)$ be the generating function for the no. of partitions of an integer n where each summand must be odd and must appear an odd no of times.

$$1 \rightarrow (x^1)^1 + (x^1)^3 + (x^1)^5 + \ldots = \frac{x}{1-x^2}$$

($\because x$ is common)

$3 \to (x^3)^0 + (x^3)^1 + (x^3)^2 + \cdots = x^3(1 + x^6 + x^{12} \cdots)$

$= \frac{x^3}{1 - x^6}$

$5 \to (x^5)^0 + (x^5)^1 + (x^5)^2 + \cdots = x^5(1 + x^{10} + x^{20} \cdots)$

$= \frac{x^5}{1 - x^{10}}$

$g(x) = \frac{x}{1-x^2} \times \frac{x^3}{1-x^6} \times \frac{x^5}{1-x^{10}}$

$= \prod_{n=1}^{\infty} \frac{x^{2k-1}}{1 - x^{2(2k-1)}}$

6. S.T. the no. of partitions of a +ve integer n where no summand appears more than twice is equal to the no. of partitions of integer n where no summand is divisible by 3.

⇒ Let $f(x)$ be the generating function for the no. of partitions of an integer n where no summand is divisible by 3.

$1 \to (x^1)^0 + (x^1)^1 + (x^1)^2 + \cdots = 1/1 - x^1$

$2 \to (x^2)^0 + (x^2)^1 + (x^2)^2 + \cdots = 1/1 - x^2$

$4 \to (x^4)^0 + (x^4)^1 + (x^4)^2 + \cdots = 1/1 - x^4$

$5 \to (x^5)^0 + (x^5)^1 + (x^5)^2 + \cdots = 1/1 - x^5$

$7 \to (x^7)^0 + (x^7)^1 + (x^7)^2 + \cdots = 1/1 - x^7$

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$$f(x) = \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^4} \times \frac{1}{1-x^5} \times \frac{1}{1-x^7} \times \cdots \rightarrow (1)$$

Let $g(x)$ be the g.f for the no. of partitions of an integer n for which no summand appears more than twice.

1 - $(x^1)^0 + (x^1)^1 + (x^1)^2 = 1 + x + x^2 = 1 - x^3 / 1 - x$

2 - $(x^2)^0 + (x^2)^1 + (x^2)^2 = 1 + x^2 + x^4 = 1 - x^6 / 1 - x^2$

3 - $(x^3)^0 + (x^3)^1 + (x^3)^2 = 1 - x^9 / 1 - x^3$

4 - $(x^4)^0 + (x^4)^1 + (x^4)^2 = 1 - x^{12} / 1 - x^4$

$$g(x) = \frac{1-x^3}{1-x} \times \frac{1-x^6}{1-x^2} \times \frac{1-x^9}{1-x^3} \times \frac{1-x^{12}}{1-x^4} \times \cdots$$

$$\therefore \frac{1}{1-x} \times \frac{1}{1-x^2} \times \frac{1}{1-x^4} \times \cdots \rightarrow (2)$$

from (1) & (2) $\therefore f(x) = g(x)$.

7. Determine the g.f for the sequence a_0, a_1, a_2, a_n where a_n is the no of partitions of the non negative integer n into. (+ve by default)

i) Even summands

ii) distinct even summands

iii) distinct odd summands

i) Let $f(x)$ be the g.f for the no. of partitions of an integer n where each summands are even.

$2 \to (x^2)^0 + (x^2)^1 + (x^2)^2 + \ldots = \frac{1}{1-x^2}$

$4 \to (x^4)^0 + (x^4)^1 + (x^4)^2 + \ldots = \frac{1}{1-x^4}$

$6 \to (x^6)^0 + (x^6)^1 + (x^6)^2 + \ldots = \frac{1}{1-x^6}$

$$f(x) = \frac{1}{1-x^2} \times \frac{1}{1-x^4} \times \frac{1}{1-x^6} \times \ldots$$

$$\therefore \prod_{k=1}^{\infty} \frac{1}{1-x^{2k}}$$

ii) Let $g(x)$ be the gf for the no. of partitions of an integer n into distinct even summands

$2 \to (x^2)^0 + (x^2)^1 = 1 + x^2$

$4 \to (x^4)^0 + (x^4)^1 = 1 + x^4$

$6 \to (x^6)^0 + (x^6)^1 = 1 + x^6$

. . . . . .

$$g(x) = (1+x^2)(1+x^4)(1+x^6) \ldots \ldots$$

$$\prod_{k=1}^{\infty} (1 + x^{2k})$$

iii) Let $h(x)$ be the gf for the no. of partitions of an integer n into distinct odd summands

$1 \to (x^1)^0 + (x^1)^1 = 1 + x$

$3 \to (x^3)^0 + (x^3)^1 = 1 + x^3$

$5 \to (x^5)^0 + (x^5)^1 = 1 + x^5$

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$$n(x) = (1+x)(1+x^3)(1+x^5) \prod_{K=1}^{\infty} (1+x^{2K-1})$$